THE UNIVERSAL THETA TWISTING A PULLBACK FROM RELATIVE PROJ

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Representation of F_d. lemma-relative-proj (carded this batch). U_d represents F_d with its universal object.
- Base change of F_d. lemma-proj-base-change (carded this batch). Lets the universal-family verification restrict to affine opens of S.
- Universal triple. lemma-proj-functor-strict (same chapter, reformulation pointer, not carded). States that (d, O_{U_d}(d), psi^d) is the universal triple over U_d.
- Affine quadruple-triple identification. lemma-relative-proj-affine (same chapter, not carded). Translates the universal triple back into the universal quadruple.

Source reference. Original lemma 01NR: sources/stacks-constructions/pinned-original-source.tex, source lines 3682–3682. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01NT: sources/stacks-constructions/pinned-original-source.tex, source lines 3837–3837. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01N9: sources/stacks-constructions/pinned-original-source.tex, source lines 2815–2815. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01NY: sources/stacks-constructions/pinned-original-source.tex, source lines 4074–4074. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01NU: sources/stacks-constructions/pinned-original-source.tex, source lines 3868–3868. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01NN: sources/stacks-constructions/pinned-original-source.tex, source lines 3580–3580. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In Situation 01NN. Let $(f : T \to S, d, \mathcal{L}, \psi)$ be a quadruple. Let $r_{d,\mathcal{L},\psi} : T \to \underline{Proj}_S(\mathcal{A})$ be the associated S-morphism. There exists an isomorphism of $\mathbf{Z}$-graded $\mathcal{O}_T$-algebras*

$$\theta : r_{d,\mathcal{L},\psi}^* \left(\bigoplus\nolimits_{n \in \mathbf{Z}} \mathcal{O}_{\underline{Proj}_S(\mathcal{A})}(nd) \right) \longrightarrow \bigoplus\nolimits_{n \in \mathbf{Z}} \mathcal{L}^{\otimes n}$$

such that the following diagram commutes

$$\begin{array}{ccc} \mathcal{A}^{(d)} & \xrightarrow{\psi} & f_* \left(\bigoplus_{n \in \mathbf{Z}} \mathcal{L}^{\otimes n} \right) \\ & {\scriptstyle \psi_{univ}} \searrow \quad \nearrow {\scriptstyle \theta} & \\ & \pi_* \left(\bigoplus_{n \geq 0} \mathcal{O}_{\underline{Proj}_S(\mathcal{A})}(nd) \right) & \end{array}$$

The commutativity of this diagram uniquely determines θ.

Proof. Note that the quadruple $(f : T \to S, d, \mathcal{L}, \psi)$ defines an element of $F_d(T)$. Let $U_d \subset \underline{\mathrm{Proj}}_S(\mathcal{A})$ be the locus where the sheaf $\mathcal{O}_{\underline{\mathrm{Proj}}_S(\mathcal{A})}(d)$ is invertible and generated by the image of $\psi_{univ} : \pi^*\mathcal{A}_d \to \mathcal{O}_{\underline{\mathrm{Proj}}_S(\mathcal{A})}(d)$. Recall that U_d represents the functor F_d, see the proof of Lemma 01NY. Hence the result will follow if we can show the quadruple $(U_d \to S, d, \mathcal{O}_{U_d}(d), \psi_{univ}|_{\mathcal{A}^{(d)}})$ is the universal family, i.e., the representing object in $F_d(U_d)$. We may do this after restricting to an affine open of S because (a) the formation of the functors F_d commutes with base change (see Lemma 01NT), and (b) the pair $(\bigoplus_{n \in \mathbf{Z}} \mathcal{O}_{\underline{\mathrm{Proj}}_S(\mathcal{A})}(n), \psi_{univ})$ is constructed by glueing over affine opens in S (see Lemma 01NR). Hence we may assume that S is affine. In this case the functor of quadruples F_d and the functor of triples F_d agree (see proof of Lemma 01NU) and moreover Lemma 01N9 shows that $(d, \mathcal{O}_{U_d}(d), \psi^d)$ is the universal triple over U_d. Going backwards through the identifications in the proof of Lemma 01NU shows that $(U_d \to S, d, \mathcal{O}_{U_d}(d), \psi_{univ}|_{\mathcal{A}^{(d)}})$ is the universal quadruple as desired. □